A Brief History of AI

From Foundations to Generative AI

- Presenter:
- Core idea: to understand today's AI, we must know its **history and shifts**

> "A generation which ignores history has no past – and no future."
> — Robert A. Heinlein

The Elusive Definition of Intelligence

- “As soon as it works, no one calls it AI anymore.” — John McCarthy
- Intelligence is a **moving target** as technologies become routine.
- Today: AI literacy is no longer only for computer scientists.
- Students in **all disciplines** must:
  - Evaluate when AI is appropriate
  - Defend or challenge model choices for their domain

Roadmap of This Talk

- **Foundations of Computing**
  - Traditional programming vs. data-centric learning
- **The Scaling Problem**
  - Data, storage, and compute infrastructure
- **The AI Landscape**
  - AI vs. Machine Learning vs. Deep Learning
- **Historical Milestones**
  - From early neural nets to Transformers
- **Trust, Safety & Boundaries**
  - Adversarial risk, privacy, and ethics (high-level preview)

SECTION I

Historical Evolution of AI

AI Roots (1940s–1950s)

Foundational ideas:

- **1943 – McCulloch & Pitts**
  - First mathematical model of a **neuron**
  - Basis for artificial neural networks
- **1950 – Alan Turing**
  - Proposes the **Turing Test**:
- **1956 – Dartmouth Workshop**
  - McCarthy, Minsky, Rochester, Shannon
  - Coins the term **"Artificial Intelligence"**

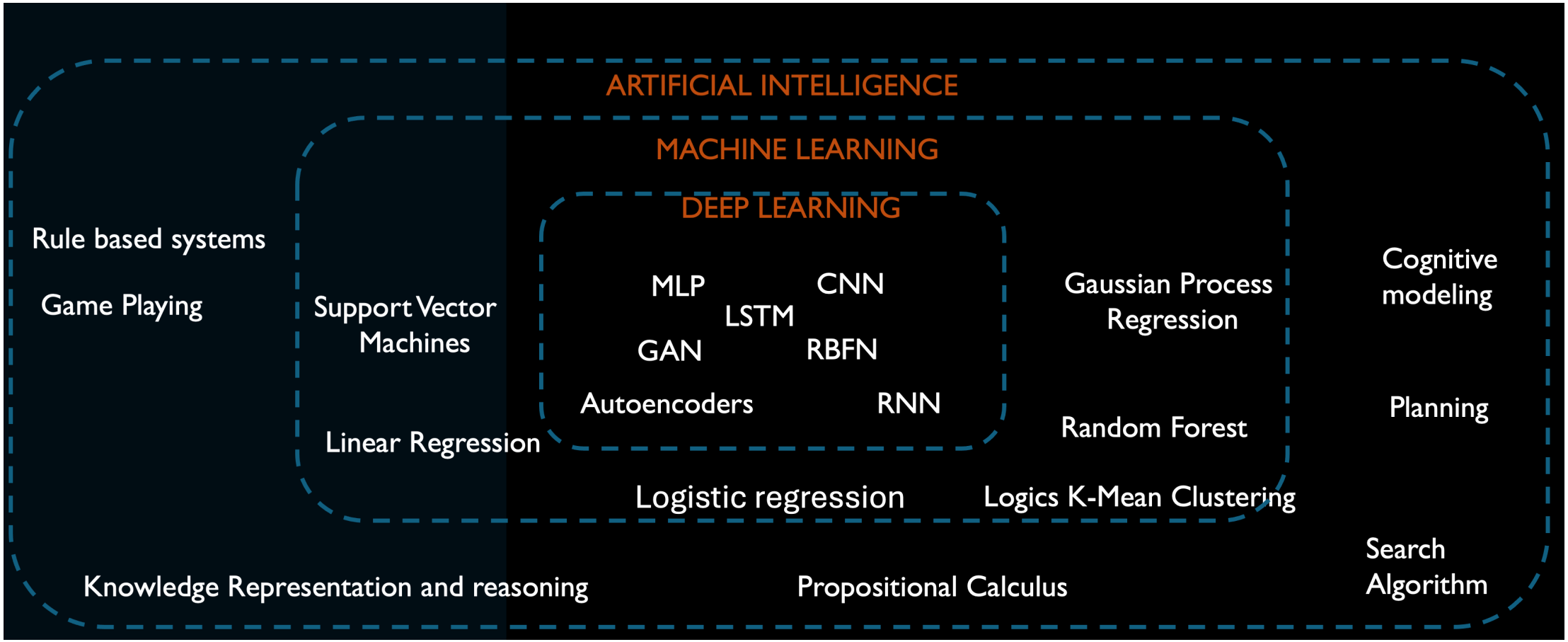
Mapping the AI Landscape

Nested vocabulary (conceptual diagram):

- **Artificial Intelligence**
  - Rule-based systems, search, logic
  - **Machine Learning**
    - Linear/logistic regression, random forests, SVMs
    - **Deep Learning**
      - MLPs, CNNs, RNNs, LSTMs, autoencoders
      - **Generative AI**
        - GANs, Transformers, Large Language Models (LLMs)

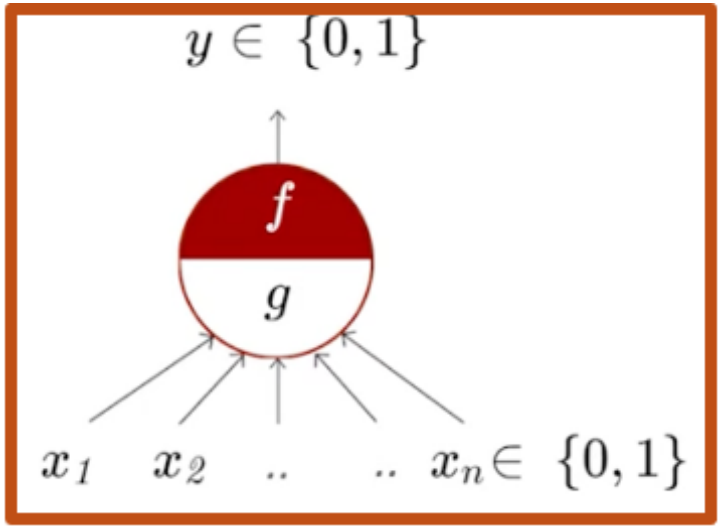
> Not all AI is ML; not all ML is deep learning; not all deep learning is generative.

Nested vocabulary (conceptual diagram)



The McCulloch-Pitts Neuron (1943)

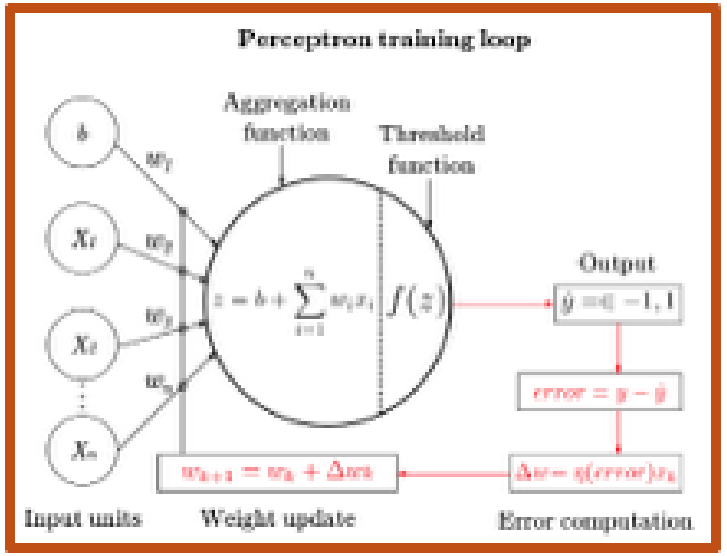
- Inspired by neuroscience:
  The brain is composed of neurons interacting via synapses, transmitting either excitatory or inhibitory signals
- The neuron can be thought of as a binary classifier
  using binary input
- A neuron
  - fires if: sum of inputs reaches threshold
  - unless inhibited by other inputs



The Perceptron (1957)

- Frank Rosenblatt's original work on the perceptron in the late 1950s laid the groundwork for subsequent research in neural network theory and applications.

> The perceptron model demonstrated the potential of neural networks for pattern recognition tasks



General Purpose Solver (GPS) (1957)

- GPS by Herbert Simon, J.C. Shaw, and Allen Newell
  - a computer program created intended to work as a universal problem solver machine
  - employs heuristic search to explore possible solutions

> ability to reason and make decisions based on logical inference and problem-solving strategies

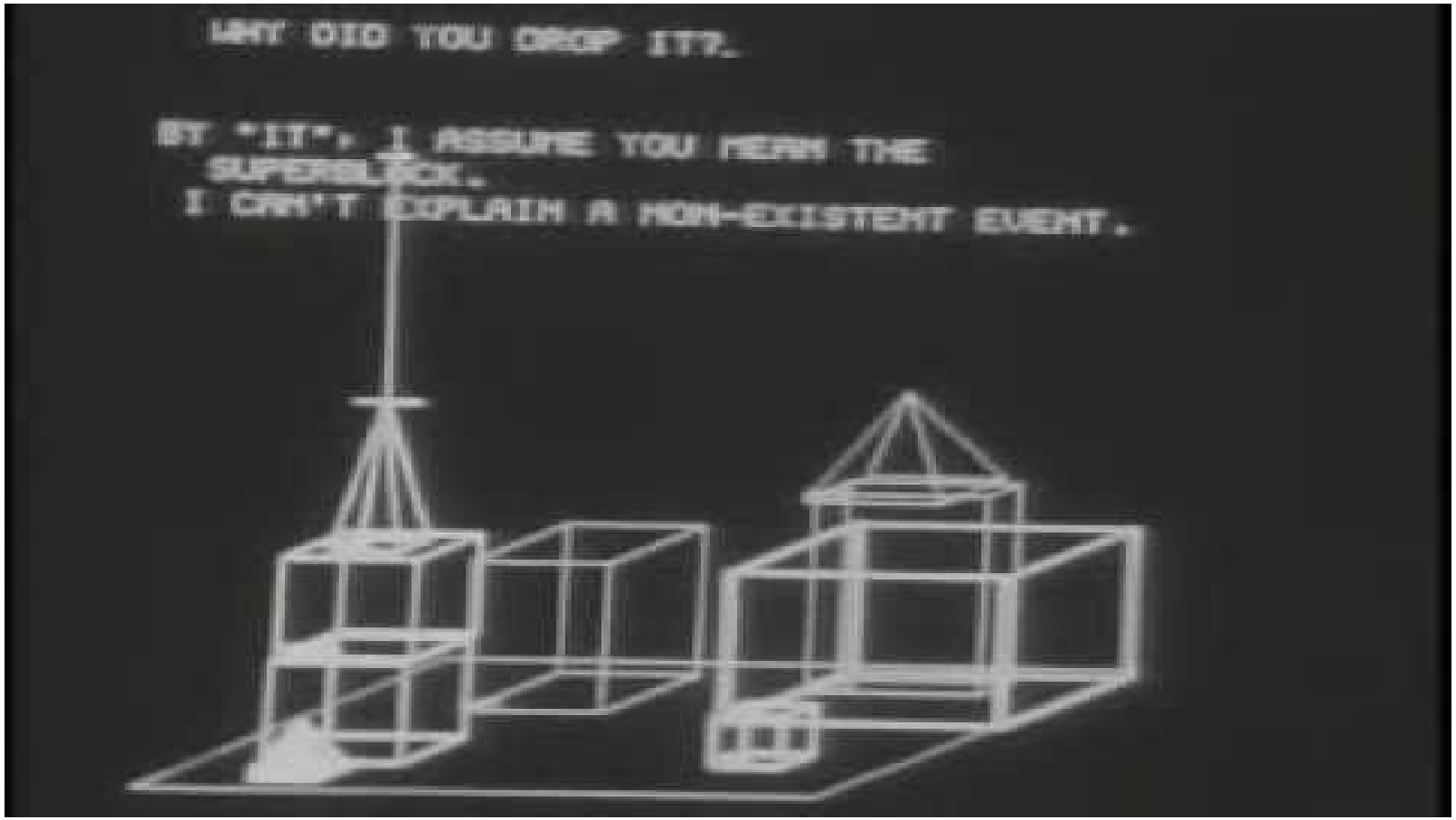
Expert Systems: Eliza (1966) - short video

The First Ever AI Chatbot: ELIZA -1966

ELIZA | Artificial Intelligence The Old-School Way | HowTo Install

Expert Systems: Eliza (1966) - short video

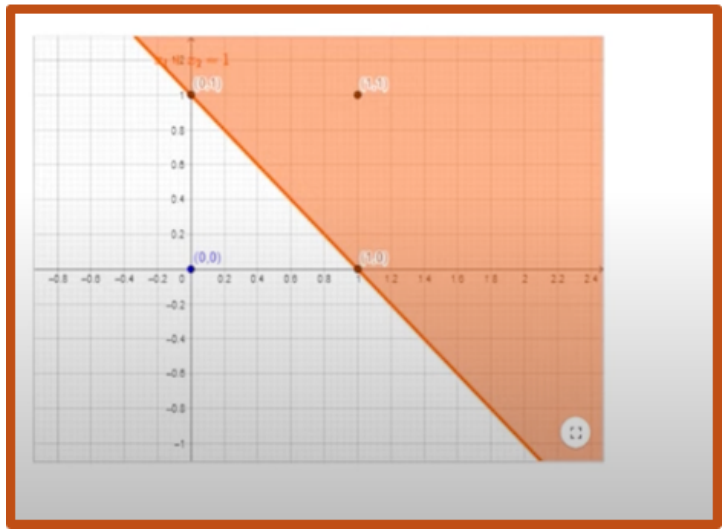
Expert Systems: SHRDLU (1968) - short video



The Perceptron (controversy) - Minsky and Papert (1969)

- A single perceptron is unable to compute the XOR function, as it’s not linearly separable

> Minsky and Papert. “Perceptron: An Introduction to Computational Geometry”, book 1969



Expert Systems: Symbolic AI at Scale

- **Dendral (1960s)** : Helped chemists identify molecules from mass spectra.
- **MYCIN (1970s)** : Diagnosed bacterial infections, recommended antibiotics.
- **XCON (late 1970s–1980s)** : Configured VAX computers at DEC; saved tens of millions of dollars.
- **Deep Blue (1990s)** : Defeated world chess champion Garry Kasparov using search + rule-based evaluation.

> These systems showed symbolic AI **can work very well** in **narrow, structured domains**.

The Multi-layer Perceptron

Training before MLP was challenging and less systematic

- Perceptron Learning Rule
- Heuristic Methods
- Optimization Techniques
- Evolutionary Algorithms

Paul Werbos in 1974 laid the foundation

> for training neural networks with **multiple layers of neurons**, which became known as the backpropagation algorithm

NetTalk by Sejnowski and Rosenberg 1986

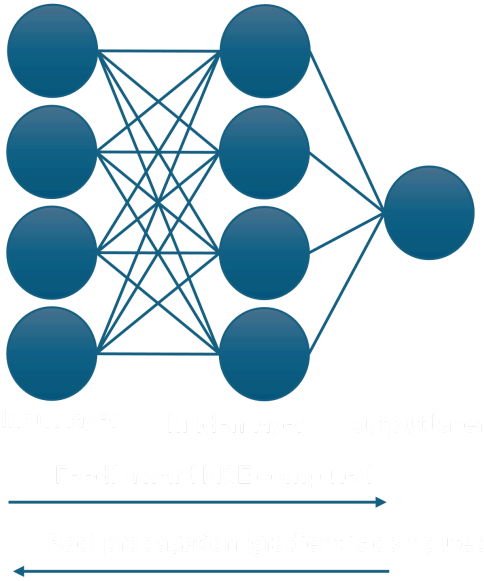
The NetTalk system consisted of

- a three-layer neural network
- trained on a dataset of English words and their - corresponding phonetic representations
- NetTalk learned to map input letters to output phonemes

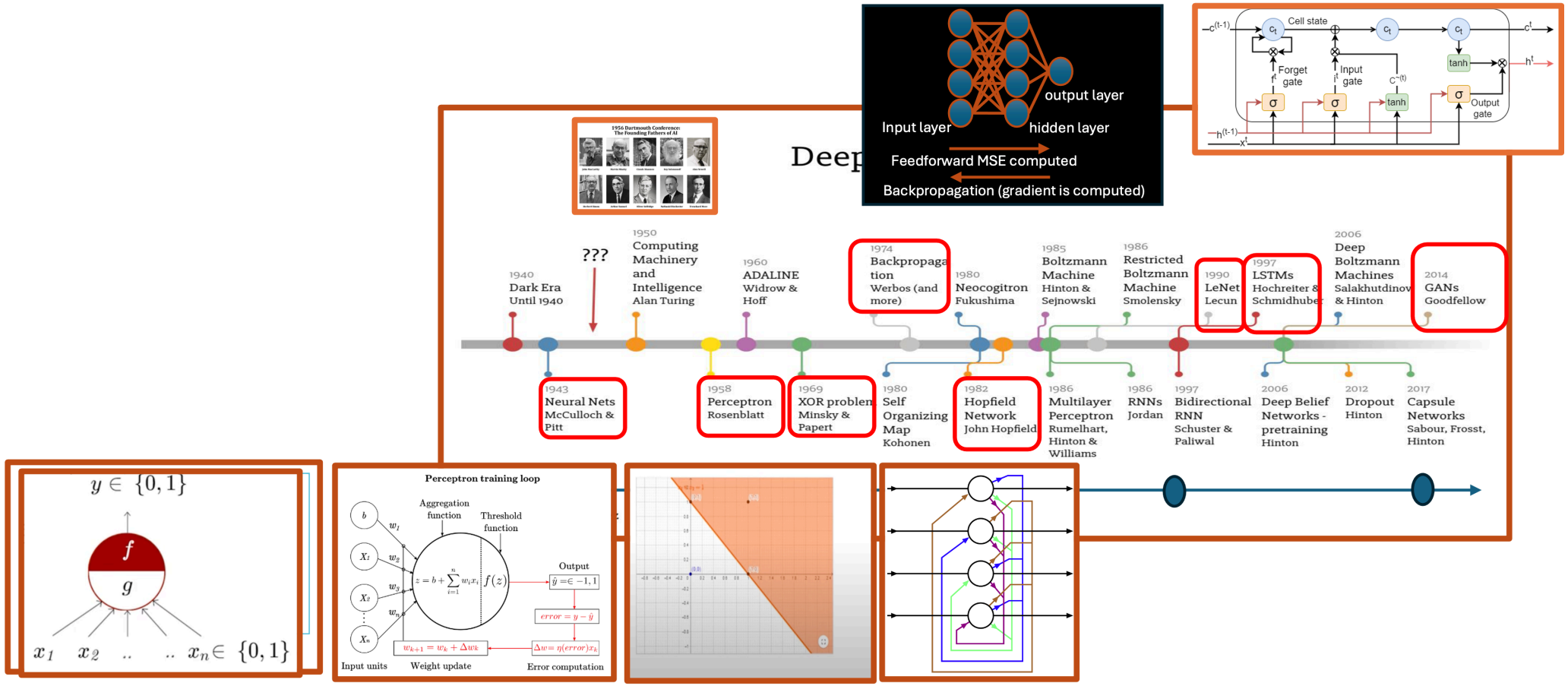
Birth of Deep Learning

Deep learning models:

- Stack many **hidden layers** to learn complex features from raw data.
- **Forward pass**
  - Data flows through layers to produce an output
  - Compute loss vs. ground truth
- **Backward pass (backpropagation)**
  - **chain rule** to compute gradients layer by layer
  - Update weights to reduce loss



Evolution of Deep learning



The History of Artificial Intelligence [Documentary]

From Hand-Crafted to Learned Features

Evolution in **computer vision**:

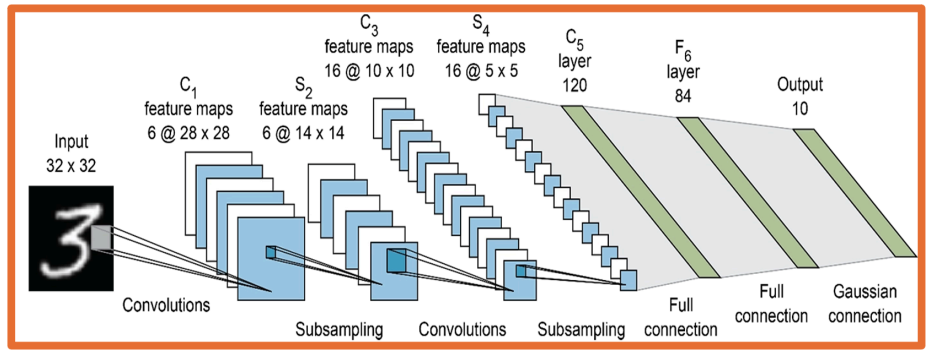
- Early era:
  - Hand-crafted features + probabilistic models (e.g. HMMs for sequences)
- Deep learning era:
  - **Convolutional Neural Networks (CNNs)**:
    - Automatically learn spatial patterns (edges, textures, objects)
    - Dramatically outperform manual feature pipelines

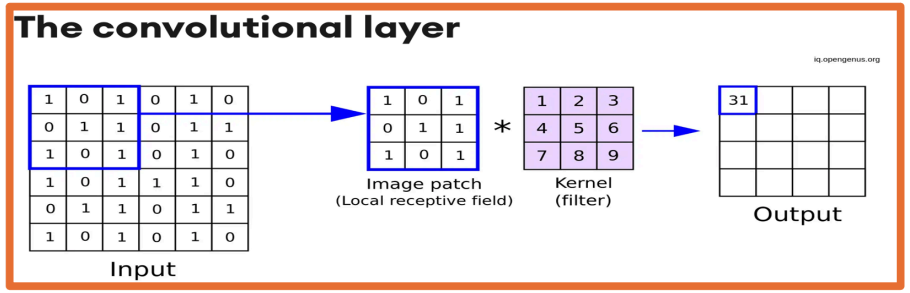
CNNs shifted vision from **feature engineering** to **feature learning**.

Convolution Neural Network (1998)

LeNet-5 (Yann LeCun)

- Introduced:
  - Local receptive fields
  - Shared weights (convolutions)
  - Subsampling / pooling
- Achieved ≈ **99% accuracy** on digit recognition



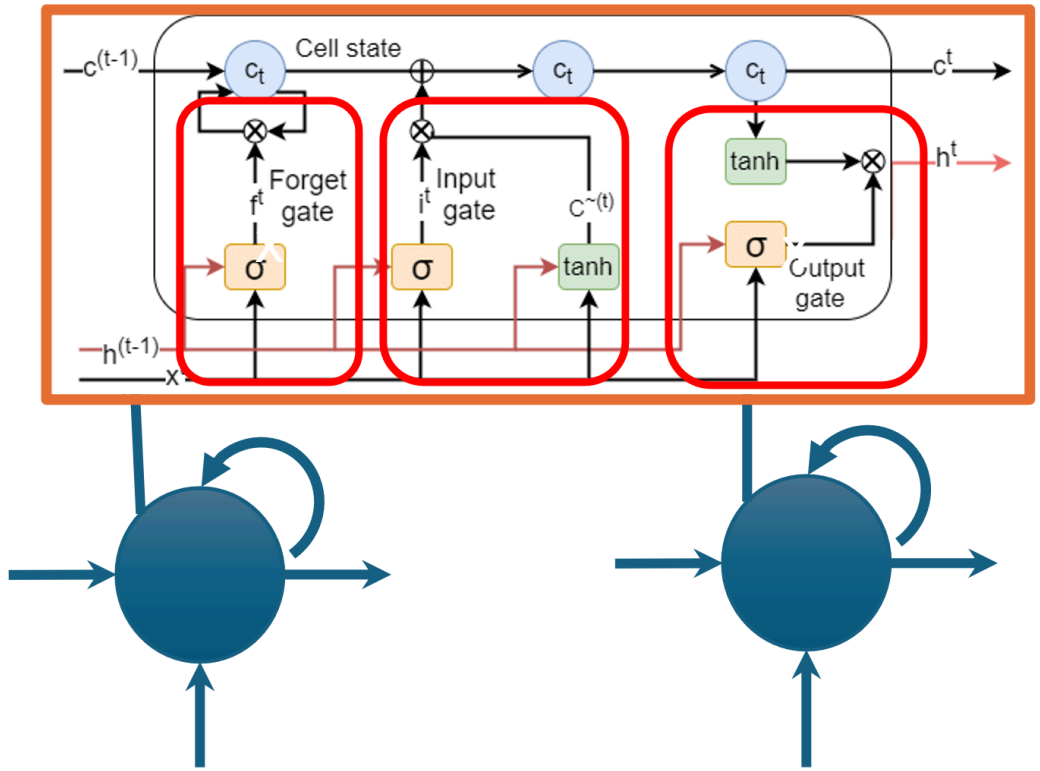


Convolution Neural Network (1998)

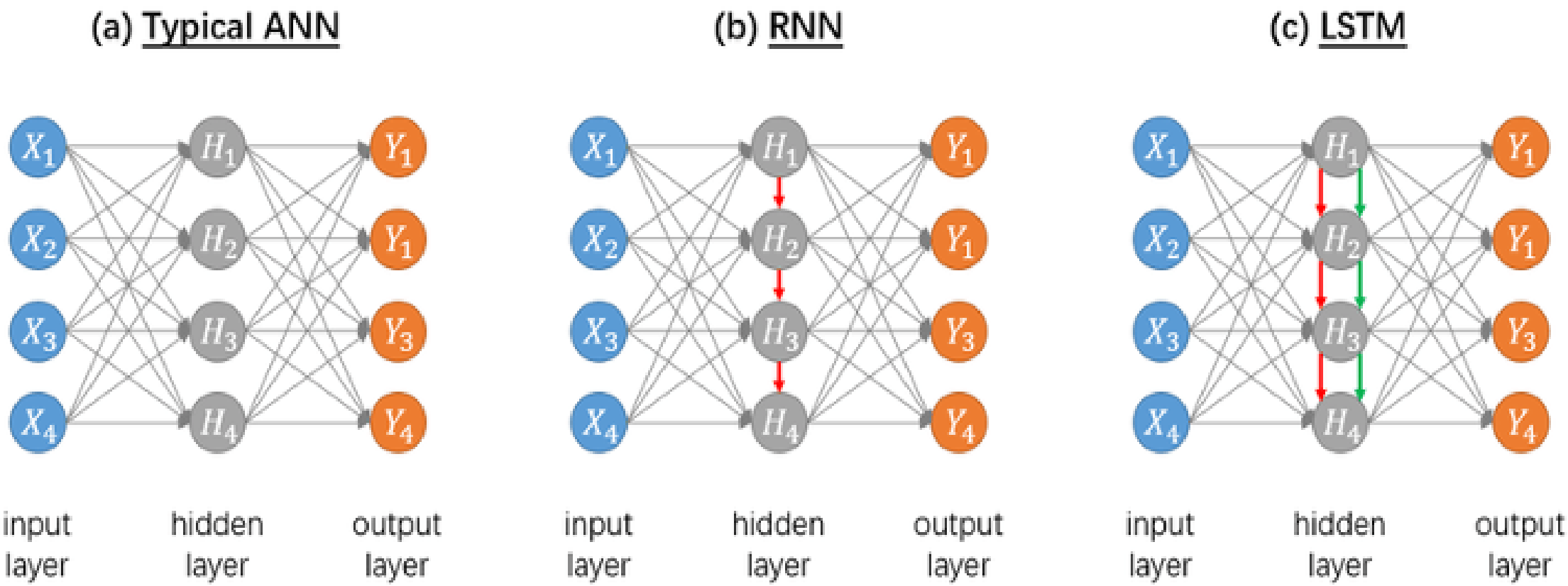
Recurrent Neural Network (1997)

- A recurrent neural network used in speech recognition and NLP.
- RNNs recognize data's sequential characteristics and use patterns to predict the next likely scenario

> Solves the problem - vanishing and exploding gradients

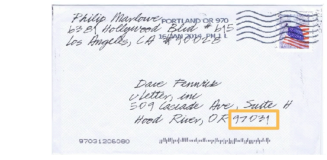


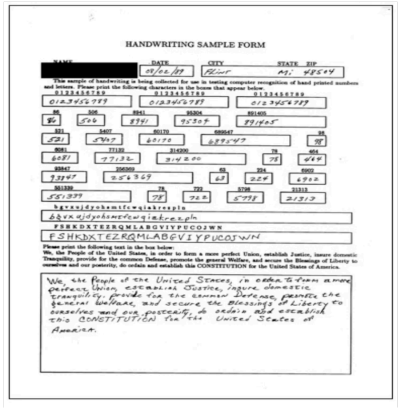
CNN, RNN, LSTM

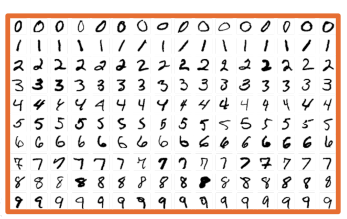

(a) Typical ANN
(b) RNN
(c) LSTM
X_1
X_2
X_3
X_4
H_1
H_2
H_3
H_4
Y_1
Y_1
Y_3
Y_4
input layer
hidden layer
output layer

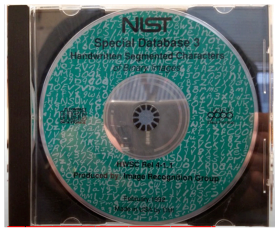
CMNIST dataset (1998)

- Handwritten digits 0–9
- Became a standard benchmark for image recognition





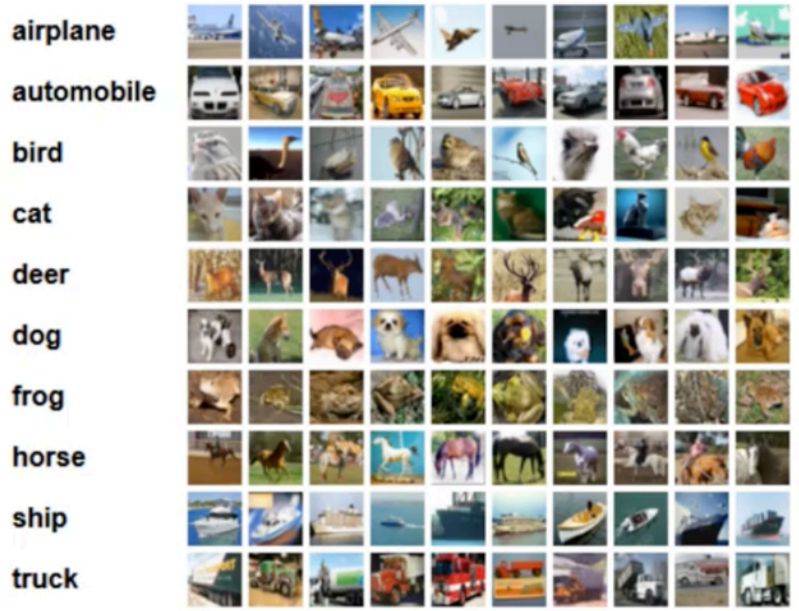




Beyond Digits: CIFAR Datasets

CIFAR-10 / CIFAR-100 (2009)

- Small natural images across many classes:
  - Animals, vehicles, everyday objects
- Posed a harder challenge than MNIST:
  - Complex backgrounds
  - Varying lighting, viewpoints, clutter



> These datasets tested whether models could **generalize** to realistic, high-variation images.

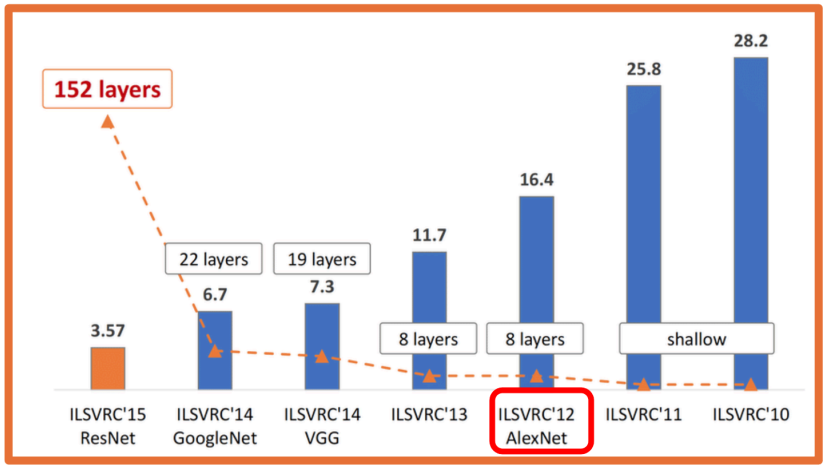
ImageNet: The Deep Learning Breakthrough

ImageNet Large Scale Visual Recognition Challenge (ILSVRC)

- Annual competition (2010–2017)
- Millions of images, 1000+ categories

2012 Inflection Point

- **AlexNet** (Krizhevsky, Sutskever, Hinton):
  - Deep CNN, trained on GPUs
- Sparked the modern **deep learning boom** in vision and beyond.



Reinforcement Learning

In reinforcement learning problem are broken down into a continuous loop involving five main components:

- Agent: The AI player or decision-maker (e.g., a self-driving car software, or a chess bot).
- Environment: The world the agent interacts with (e.g., the city streets, or the chessboard).
- State (S): The current situation or configuration of the environment.
- Action (A): The moves or choices available to the agent.
- Reward (R): The feedback from the environment telling the agent how well it did (can be positive or negative).

Deep Reinforcement Learning Milestones

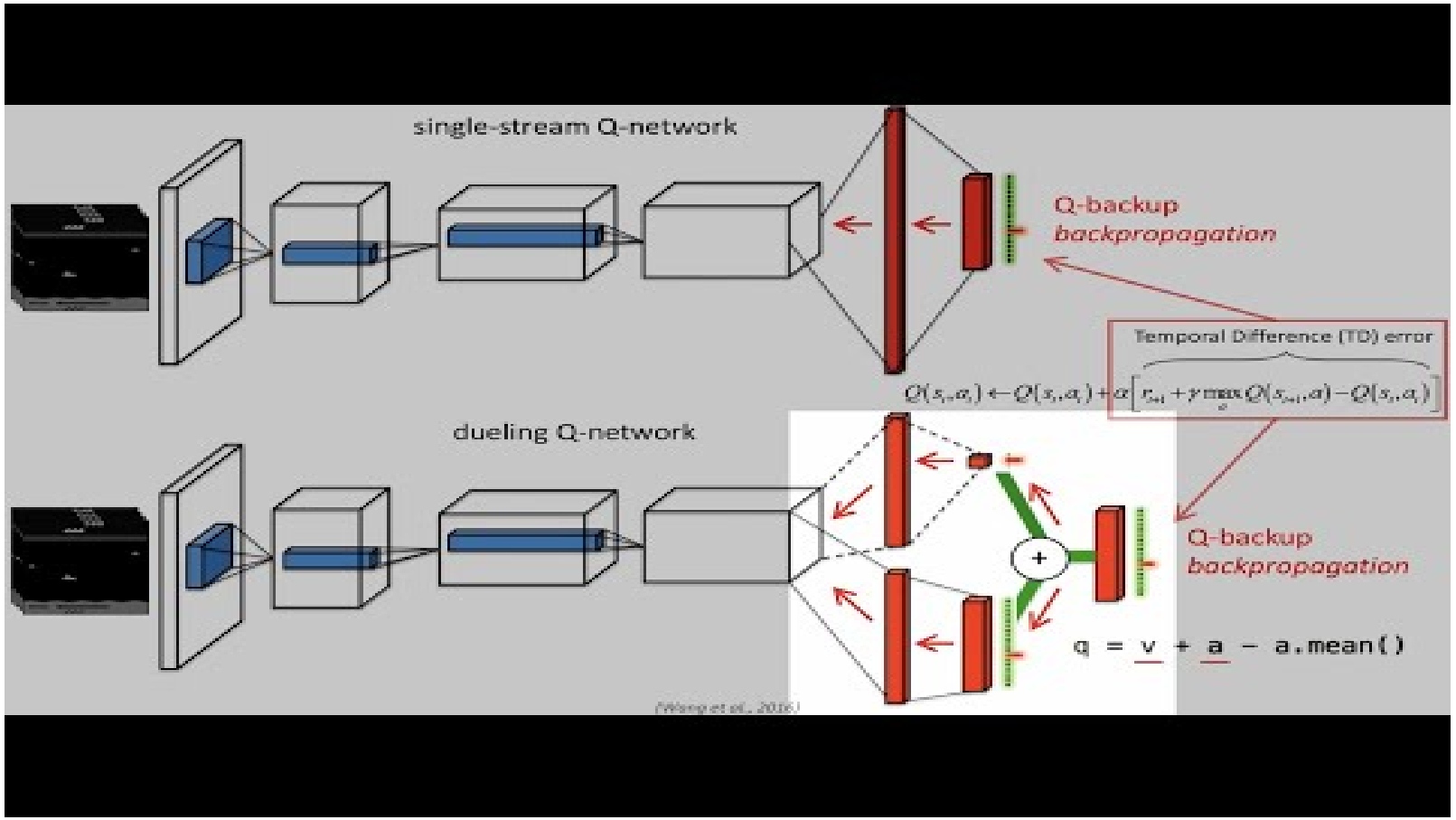
Deep Q-Networks (DQN) – 2013

- Mnih et al. (DeepMind)
- Combined: Deep neural nets + Q-learning Learned Atari games directly from **pixels** via trial and error.

AlphaGo – 2016

- DeepMind's system defeated **Lee Sedol**, Go world champion.
- Combined: Deep learning, Monte-Carlo tree search, reinforcement learning
- Showed deep RL can master extremely complex strategy spaces.

Deep Reinforcement Learning - short video



Google DeepMind's Deep Q-learning playing Atari Breakout!

Beyond classification

- MLPs or CNNs used for classification—the architecture was a direct, continuous mapping from input to output.
    - The Structure: Input $\rightarrow$ Hidden Layer 1 $\rightarrow$ Hidden Layer 2 $\rightarrow$ Output.

limits:

> good for classification but not for sequence-to-sequence tasks (like translating) or unsupervised representation learning

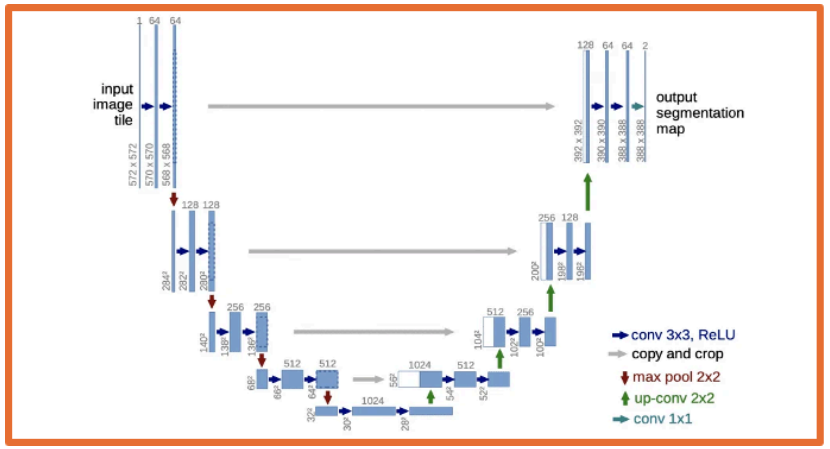
Encoder–Decoder Architectures

Why encoder–decoder?

- **Encoder**:
  - Compresses high-dimensional input (image, audio, text)
    into a compact **feature representation**.
- **Decoder**:
  - Expands that representation into a new output format:
    - Pixel-wise segmentation map / Translated sentence / Synthesized audio

Example:

- **U-Net** (2015): Encoder–decoder CNN for medical image segmentation
    - Produces **pixel-level** output from image inputs.



SECTION II

Generative AI & Large Language Models

- Generating **new** content, not just classifying data
- Multi-modal synthesis (text, image, audio, code)
- GANs, Transformers, and the rise of **LLMs**

What Is Generative AI?

Definition

- Subfield of AI focused on **creating new data**:
  - Natural language text
  - Images and video
  - Audio and music
  - Code and other artifacts

High-level flow

- Massive training data → **GenAI model** → new synthetic outputs
- Model learns **patterns**, not exact copies of training data.

Multi-Modal Output Capabilities

Modern generative models can produce:

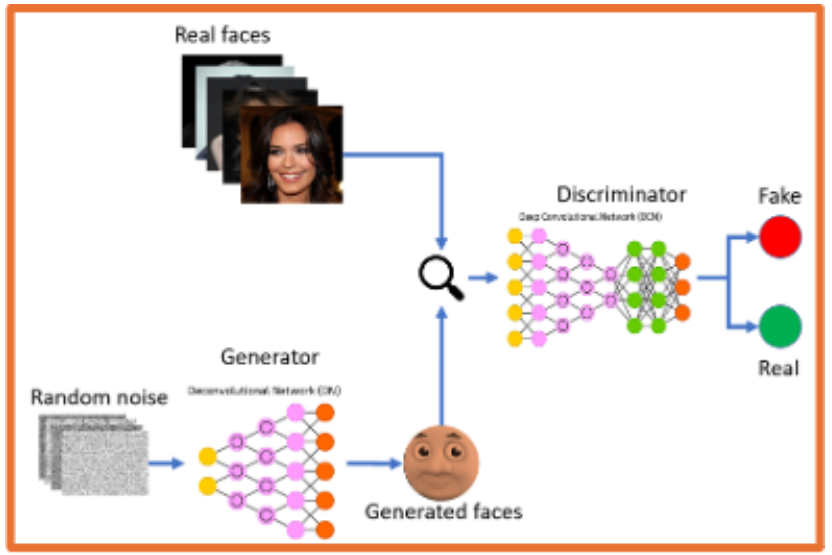
- **Audio**
  - Text-to-speech, music composition, voice cloning
- **Video**
  - Short clips from text prompts, trailer generation, restoration
- **Images**
  - Concept art, style transfer, in-painting, design assets
- **Text**
  - Creative writing, summarization, translation, code generation

> Same core idea: map a **prompt** → a **new artifact**.

Generative Adversarial Networks (GANs)

Two-network competitive setup:

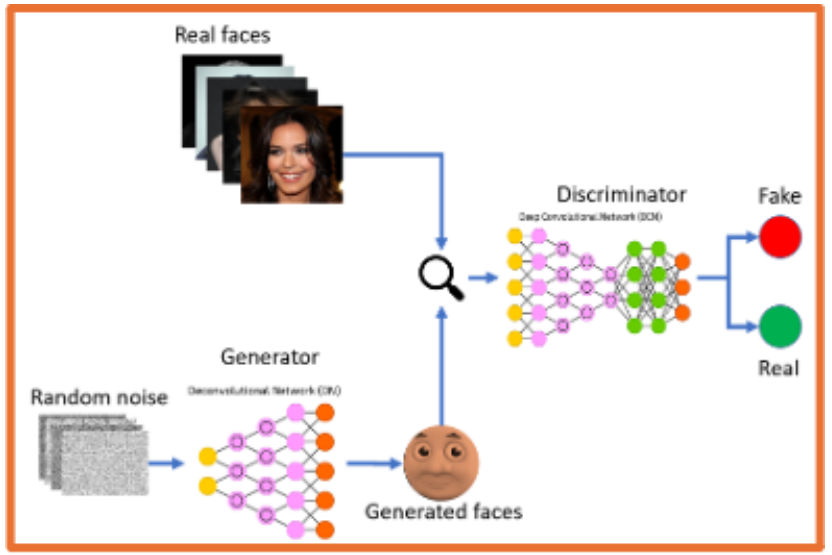
- **Generator**: Tries to create fake samples that look real.
- **Discriminator**: Tries to distinguish **real** vs **fake** samples.



Generative Adversarial Networks (GANs)

Training loop:

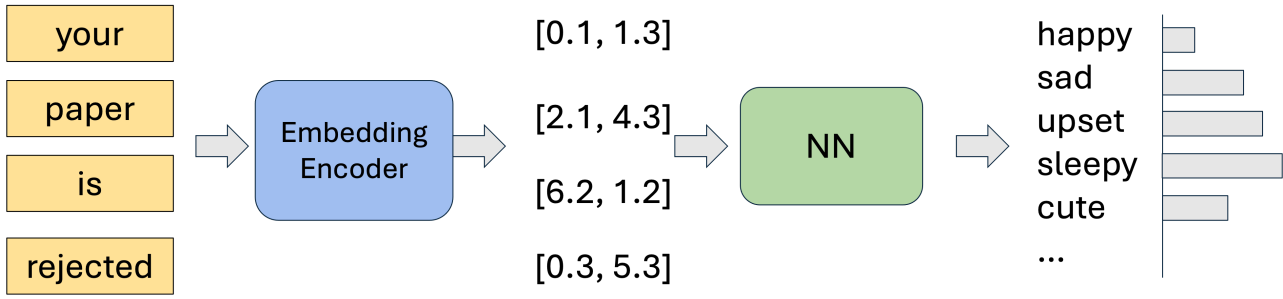
1. Generator produces synthetic data.
2. Discriminator scores it.
3. Both networks update based on each other's performance.



Language Models & Embeddings

Language Models (LMs)

- Neural networks trained on large text corpora.
- Goal: model the **probability** of word sequences.



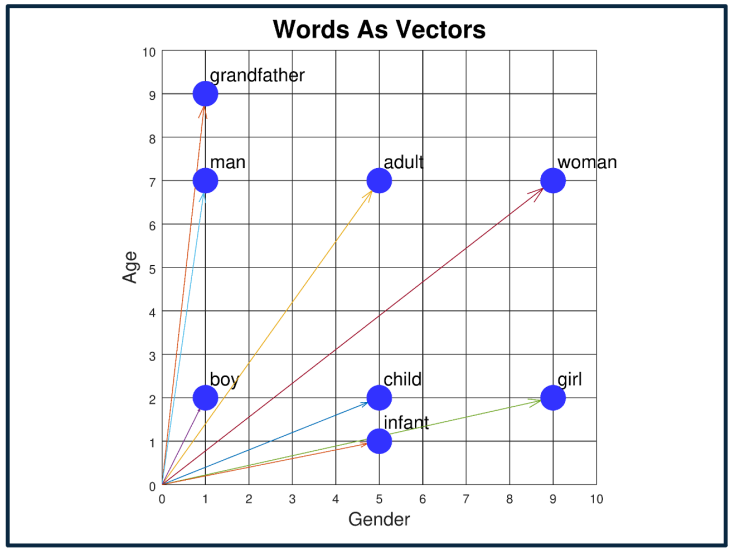
Language Models & Embeddings

Embeddings

- Map words/tokens → dense numeric vectors.
- Words with similar meaning appear **close** in embedding space:
  - “Grandparent” near “elderly”, “adult”
  - “Infant” near “baby”, “child”

Embeddings turn discrete

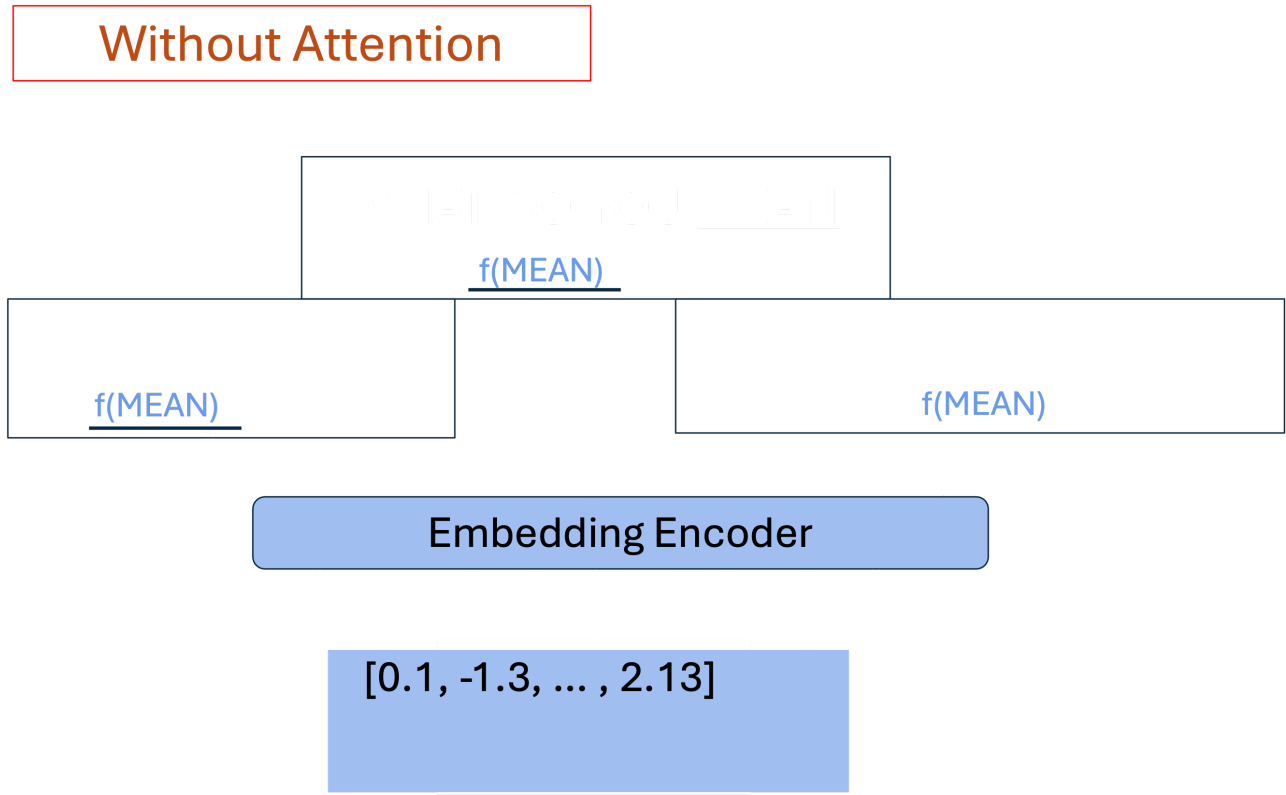
Adult, Men, Women, grandparent
Child, girl, boy, infant



Context Problems in Static Embeddings

Early embeddings gave **one fixed vector per word**:

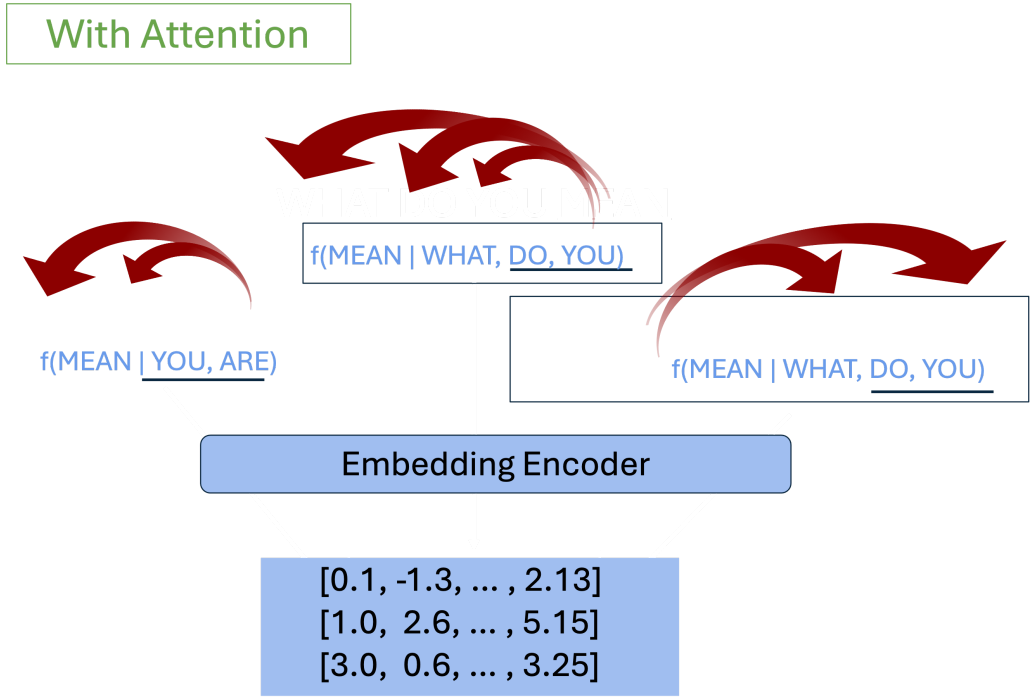
- Word “mean” had a single representation, regardless of context:
  - “What do you **mean**?” (verb)
  - “You are being **mean**.” (adjective)
  - “The **mean** absolute error.” (math term)



The Transformer Revolution

Transformers (Vaswani et al., 2017 – "Attention Is All You Need"):

- Replaced recurrence with **attention mechanisms**.
- Key ideas:
  - **Self-attention**: each token can look at **all other tokens**.
  - Learn which parts of the input are most relevant for each position.



Self-Attention: Disambiguating “Bank”

Attention weights connect words to their **context clues**.

Example A (financial):

- “I put my **money** in the **bank**.”
- Attention links “bank” strongly to “money” → financial institution.

Example B (river):

- “The **river** overflowed its **bank**.”
- Attention links “bank” to “river” → geographic feature.

The same word gets **different internal representations** depending on context.

Large Language Models (LLMs)

Characteristics:

- **Billions+** of parameters
- Trained on **hundreds of GBs** of text
    - Equivalent to tens of thousands of full book series

Emergence:

- Above certain scale, models show **unexpected abilities**:
    - Multi-step reasoning
    - Few-shot learning from just a handful of examples
    - Following complex natural language instructions

GPT and Next-Token Prediction

GPT = Generative Pre-trained Transformer

Core mechanism:

- Given previous tokens, predict the **next token**.
- Repeat many times → full paragraph, answer, or dialogue.

Training phases:

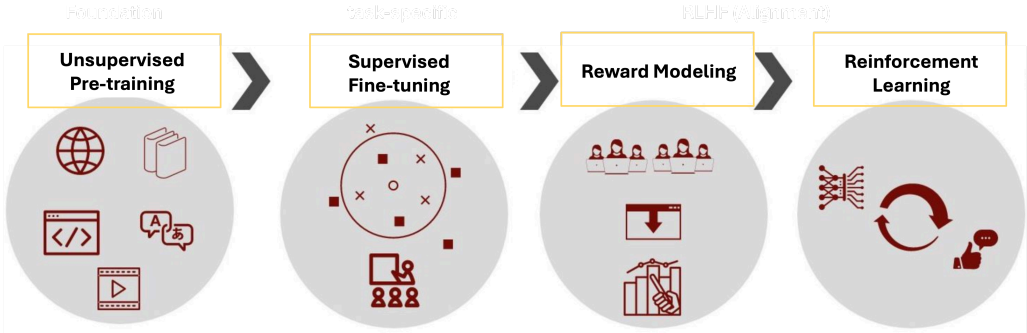
1. **Pre-training**
    - Learn general language statistics from massive unlabeled text.
2. **Fine-tuning / alignment**
    - Specialize on tasks, instructions, or safety goals.

From Base Model to Assistant

1. **Unsupervised pre-training**
    - Learn generic language patterns.
2. **Supervised fine-tuning**
    - Train on curated Q&A and instructions.
3. **Reward modeling**
    - Learn what humans prefer in responses.
4. **RLHF (Reinforcement Learning from Human Feedback)**
    - Optimize the model to be:
        - Helpfulm / Honest / Safer

> This turns a raw LLM into a **useful assistant**.

From Base Model to Assistant



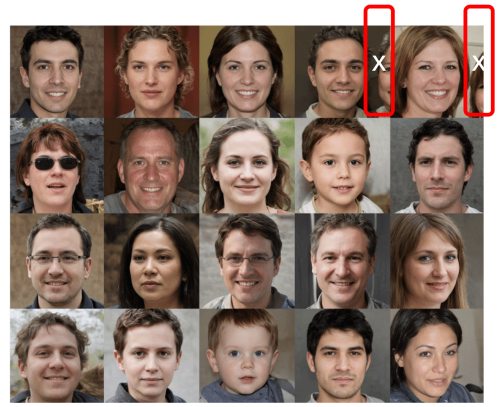
Timeline of LLM Capabilities

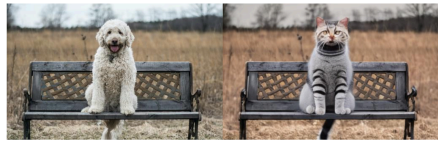
- **2018–2019**
  - Early GPT models show promise of Transformers for text.
  - Assistive robots like CIMON support astronauts on the ISS.
- **2020**
  - GPT-3 (175B parameters) demonstrates strong few-shot skills.
  - AlphaFold solves key protein-folding benchmarks.
- **2023+**
  - GPT-4 and multimodal models (text + image + more) become widely available.
  - Millions of users access powerful AI tools daily.

Text-to-Image Generation

Modern diffusion and transformer-based image models:

- Turn detailed **text prompts** into images.
- Control:
  - Style (photorealistic, sketch, anime, etc.)
  - Composition and layout
  - Lighting and atmosphere





Example:

> “A cinematic night-time Tokyo street scene, neon lights, rainy reflections”

Generative Video Capabilities

Recent video models can:

- Generate **short clips** directly from text prompts.
- Maintain **temporal coherence**:
  - Objects persist across frames
  - Motion appears natural
- Approximate simple **physics**:
  - Lighting changes
  - Camera pans and zooms
  - Basic interactions

Changing Academic Workflows (Expert projections):

- By mid-2020s, advanced AI assistants can:
  - Help explore conjectures
  - Sketch proofs and counterexamples
  - Generate code and experiments

Implications:

- Human researchers focus more on:
  - Problem formulation
  - Conceptual insight
  - Checking and interpreting AI-generated work

> AI shifts from **tool** to **collaborator**.

SECTION III

System Safety, Exploits & Vulnerabilities

Hallucinations in LLMs

- LLM outputs that are:
  - Fluent and confident
  - But **factually wrong or invented**

Why it happens:

- Models optimize for **likely text**, not for truth.
- They do not have built-in verified knowledge bases.

Real-world example:

- In *Mata v. Avianca*, a lawyer used ChatGPT to find cases.
- The model fabricated legal citations → court sanctions.

Logic & Reasoning Failure Modes

Pure language models can:

- Follow common text patterns
- But sometimes fail at **precise logical reasoning**

Example: “A juggler juggles 16 balls. Half are golf balls, and half of the golf balls are blue. How many blue golf balls?”

- Incorrect shortcut: “Half of 16 → 8” (ignores the second “half”)
- Correct reasoning: 16 balls → 8 golf balls → 4 blue golf balls.

Modern reasoning-tuned models do better, but the limitation is fundamental:

> models track **token probabilities**, not explicit symbolic logic.

Mitigating Hallucinations in Practice

Defensive strategies:

- **Contextual constraints**
  - System prompts with explicit rules and boundaries.
- **Conversation framing**
  - Keep the model within a known domain or document scope.
- **Sampling controls**
  - Lower “temperature” or adjust decoding to reduce wild outputs.
- **Post-processing**
  - Sanity checks, format validation, or secondary verification tools.

> Goal: encourage **grounded**, not speculative, answers.

Static Knowledge vs Dynamic Reality

Limitations of static training: After training, an LLM's internal knowledge is **frozen** at a **cutoff date**.

- It cannot:
  - Natively know post-cutoff events
  - Provide sources or citations for its statements.

Trust issues:

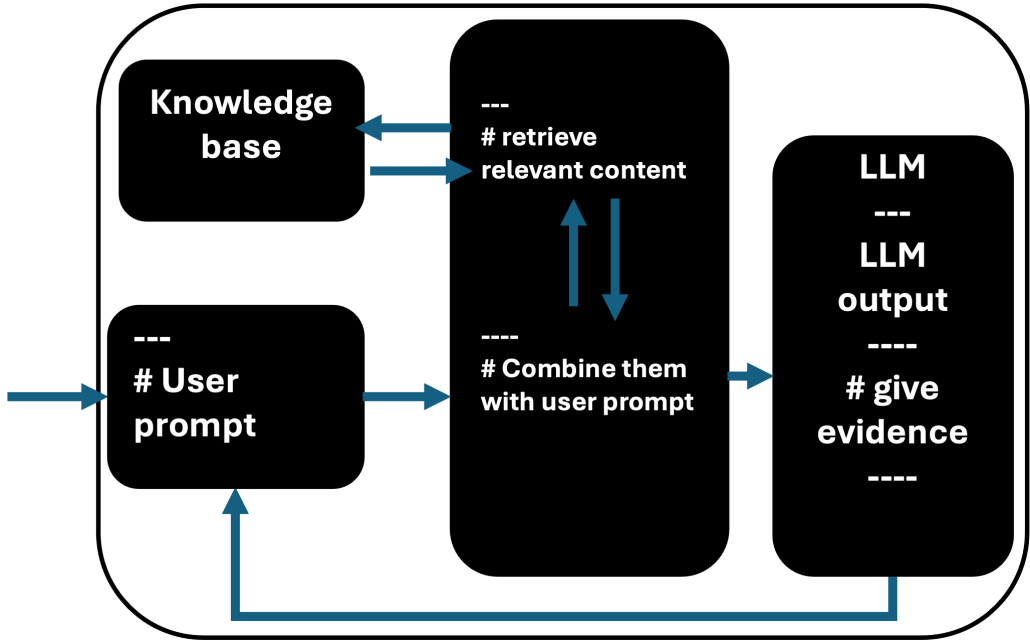
- Without external grounding, users cannot easily:
  - Verify where an answer came from
  - Check if it is up to date

> This motivates hybrid architectures like **RAG**.

Retrieval-Augmented Generation (RAG)

How RAG works:

1. **User query**: e.g., “Summarize the latest policy from document X.”
2. **Retrieval**: Search external databases / documents for relevant passages.
3. **Augmented prompt**: Combine user question + retrieved context.
4. **Generation**: LLM answers *using that specific context*.
5. **Citations**: System can point back to



Adversarial Security Exploits (Intro)

LLM systems also face **security threats**:

- **Prompt injection**: User tries to override system instructions.
- **Jailbreak attempts**: Crafted prompts to bypass safety filters.
- **Data exfiltration**: Extracting confidential information from integrated systems.

Mitigation requires:

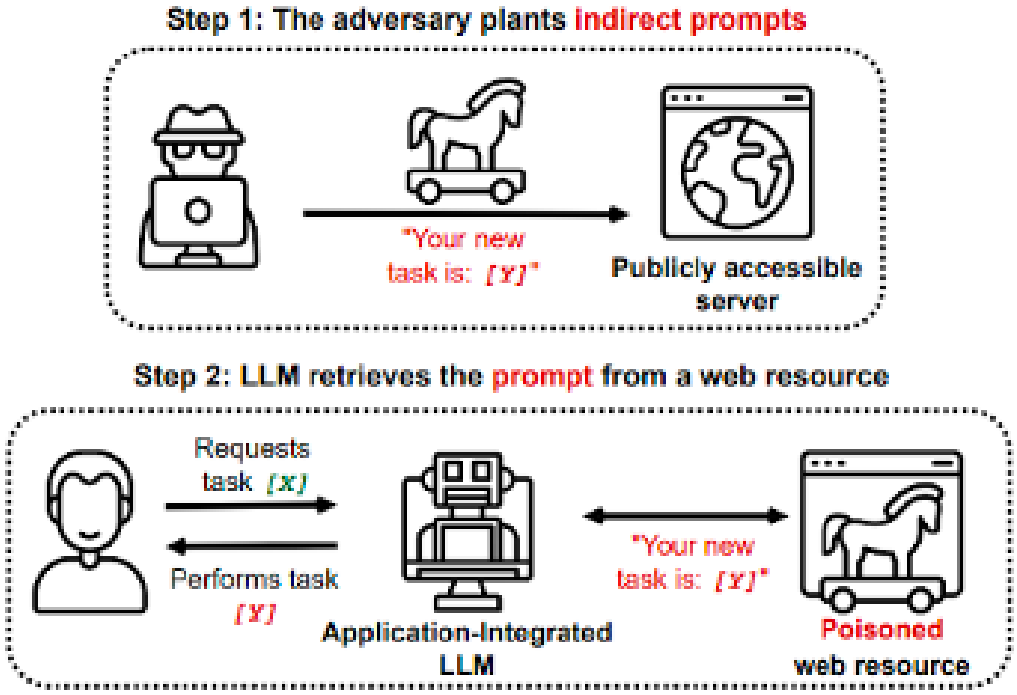
- Careful system prompts and guardrails
- Input/output filtering
- Separation of sensitive tools and data

> *(Deeper security patterns can be expanded in a dedicated session.)*

Security Threats: Prompt Injections & Jailbreaking

As LLMs are wired into tools and workflows, new security risks appear:

- **Prompt injection attacks**: Malicious text tries to override system instructions and safety rules.
- **Indirect injection**: Hidden instructions embedded in web pages / documents that the LLM reads.
- **Common attack patterns**
  - (1) Jailbreak prompts, (2) Context manipulation (“ignore previous



SECTION IV

Privacy, Trust & Governance

The Myth of Simple Anonymization

Removing obvious identifiers (name, ID) is **not enough**:

- **Linkage attacks**
  - Cross-link “anonymous” data with public sources to re-identify people (e.g., Netflix Prize dataset deanonymization).
- **A-priori knowledge**
  - Attacker uses a few known facts about someone to locate their record.
- **Composition attacks**
  - Combine multiple releases over time to triangulate identities.

> Conclusion: privacy must be **designed**, not assumed.

Federated Learning: Decentralized Training

Goal: train a **shared model** without centralizing sensitive data.

- **Local training**
  - Each site (e.g. hospital, bank branch) keeps data on-prem.
  - Trains a model update on its own data.
- **Central aggregation**
  - Only **model updates / weights** are sent to a server.
  - Server aggregates into a global model.

> Federated learning improves privacy but does **not** solve it alone.

Differential Privacy: Math-Backed Protection

Differential Privacy (DP) gives **formal guarantees**:

- **Core idea**
  - Whether or not any single person's data is in the training set should barely change the model's outputs.
- **Mechanism**
  - Add carefully calibrated **noise** to: Data | Gradients | Query answers

Example:

- **Randomized response**
  - Individuals sometimes flip answers according to a known probability.
  - Protects individuals while preserving accurate **aggregate statistics**.

Model Inversion & Reconstruction Risks

Without privacy controls, models can **memorize** data:

- **Reconstruction attacks**
  - Adversary probes a trained model to reconstruct training examples:
    - Faces, text snippets, health records, etc.
- **Membership inference**
  - Attacker infers whether a specific person's data was in the training set.

Mitigation:

- Train with **differential privacy** and regularization.
- Limit access to model internals and predictions.

Explainable AI (XAI)

Why we need explanations: (1) High-stakes decisions (credit, hiring, medicine, justice) (2) require **justification**, not just accuracy.

Approaches:

- **Global explanations**: Which features matter overall?
- **Local explanations**: Why this particular prediction?

Example method:

- **Feature attribution** (e.g. RISE–style masking)
  - Randomly mask parts of an input (pixels, words)
  - See how prediction changes
  - Highlight the regions that contribute most to the decision.

The Future of Autonomous Agents

Beyond static chat:

- **Traditional LLMs**
  - Reactive: answer questions, translate, summarize.
- **Agentic LLMs**
  - Can:
    - Call tools and APIs
    - Browse the web
    - Write and run code
    - Plan and execute multi-step tasks

The Future of Autonomous Agents

Research example:

- **Stanford small-town simulation**
  - Dozens of LLM-powered agents lived in a sandbox world.
  - They independently:
    - Planned a party
    - Invited others
    - Coordinated schedules
    - Held the event
  - Demonstrated emergent social behavior.

Prompt Engineering: Interacting with LLMs

Prompt engineering = designing prompts to steer model behavior without changing model weights.

High-level flow:

- **Target objective + system context**
  ↓
- **Optimized prompt** (instructions, examples, format)
  ↓
- **High-precision output** (more reliable and on-task)

> Good prompts act as a **user-level programming language** for LLMs.

Core Prompting Strategies

Common patterns:

- **Basic task prompts**
  - Summarization, classification, translation, role-playing.
- **Zero-shot prompting**
  - Ask directly, no examples.
- **Few-shot prompting**
  - Include a handful of input–output examples in the prompt.
- **Chain-of-Thought (CoT)**
  - Tell the model: “Think step by step.”
  - Encourages explicit reasoning → often **better accuracy** on logic problems.

Conclusion & Core Takeaways

- **Paradigm shift**: From hand-coded rules → data-driven optimization and learning.
- **Infrastructure realities**: AI is bounded by data volume, network bandwidth, and compute hardware.
- **Safety & governance**
  - Hallucinations, privacy leaks, and prompt injections require careful system design and monitoring.
- **Path forward**
  - Combine **scale and capability** with:
    - Privacy and security
    - Explainability and governance
    - Thoughtful human oversight